

COORDONNEES POLAIRES

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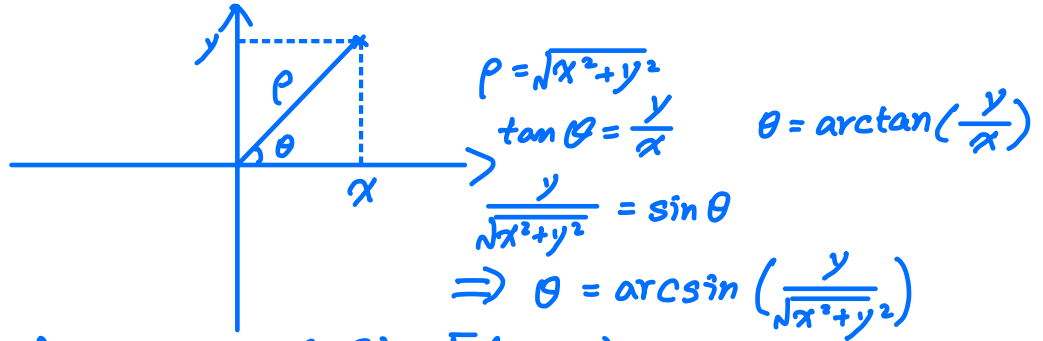
Soit le passage en coordonnées polaires défini par $(x, y) = (r \cos \theta, r \sin \theta)$.

Soit une fonction $f = f(x, y)$ définie sur $R \times R - \{0, 0\}$. On supposera que f est de classe C^2 .

Soit $F(r, \theta) = f(r \cos \theta, r \sin \theta)$. Les formules suivantes sont rappelées :

$$\frac{\partial r}{\partial x} = \cos \theta, \quad \frac{\partial r}{\partial y} = \sin \theta \quad \text{et} \quad \frac{\partial \theta}{\partial x} = -\left(\frac{\sin \theta}{r}\right), \quad \frac{\partial \theta}{\partial y} = \frac{\cos \theta}{r}$$

1. exprimez le gradient de $f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right)$ en fonction de $\frac{\partial F}{\partial r}, \frac{\partial F}{\partial \theta}$
2. exprimez le laplacien de f en fonction de $\frac{\partial^2 F}{\partial^2 r}, \frac{\partial^2 F}{\partial r \partial \theta}, \frac{\partial^2 F}{\partial^2 \theta}$



$$1. f(x, y) = f(x(r, \theta), y(r, \theta)) = f(r \cos \theta, r \sin \theta) = F(r, \theta)$$

$$\begin{aligned} \frac{\partial f}{\partial x} &= \frac{\partial F}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial F}{\partial \theta} \frac{\partial \theta}{\partial x} \\ &= \cos \theta \frac{\partial F}{\partial r} - \frac{\sin \theta}{r} \frac{\partial F}{\partial \theta} \end{aligned}$$

$$\begin{aligned} \frac{\partial f}{\partial y} &= \frac{\partial F}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial F}{\partial \theta} \frac{\partial \theta}{\partial y} \\ &= \sin \theta \frac{\partial F}{\partial r} + \frac{\cos \theta}{r} \frac{\partial F}{\partial \theta} \end{aligned} \quad (\cos(\underline{x}))'$$

$$\vec{\nabla} f(x, y) = \left(\cos \theta \frac{\partial F}{\partial r} - \frac{\sin \theta}{r} \frac{\partial F}{\partial \theta}, \sin \theta \frac{\partial F}{\partial r} + \frac{\cos \theta}{r} \frac{\partial F}{\partial \theta} \right)$$

$$\begin{aligned} 2. \frac{\partial^2 f}{\partial^2 x} &= \frac{\partial}{\partial x} \left(\cos \theta \frac{\partial F}{\partial r} - \frac{\sin \theta}{r} \frac{\partial F}{\partial \theta} \right) \\ &= \frac{\partial (\cos \theta)}{\partial x} \frac{\partial F}{\partial r} + \cos \theta \frac{\partial}{\partial x} \left(\frac{\partial F}{\partial r} \right) + \frac{\partial}{\partial x} \left(-\frac{\sin \theta}{r} \right) \frac{\partial F}{\partial \theta} + \left(-\frac{\sin \theta}{r} \right) \frac{\partial}{\partial x} \left(\frac{\partial F}{\partial \theta} \right) \\ &= -\sin \theta \cdot \frac{\partial \theta}{\partial x} \frac{\partial F}{\partial r} + \cos \theta \left(\frac{\partial^2 F}{\partial^2 r} \frac{\partial r}{\partial x} + \frac{\partial^2 F}{\partial r \partial \theta} \frac{\partial \theta}{\partial x} \right) - \frac{\partial (\sin \theta)}{\partial x} \frac{1}{r} \frac{\partial F}{\partial \theta} - \sin \theta \frac{\partial (\frac{1}{r})}{\partial x} \frac{\partial F}{\partial \theta} + \\ &= \frac{\sin^2 \theta}{r} \frac{\partial F}{\partial r} + \cos^2 \theta \frac{\partial^2 F}{\partial^2 r} - \frac{\sin \theta \cos \theta}{r} \frac{\partial^2 F}{\partial r \partial \theta} - \frac{2 \sin \theta \cos \theta}{r^2} \frac{\partial F}{\partial \theta} - \frac{\sin \theta}{r} \left(\cos \theta \frac{\partial^2 F}{\partial r \partial \theta} - \frac{\sin \theta}{r} \frac{\partial^2 F}{\partial^2 \theta} \right) \\ &= \cos^2 \theta \frac{\partial^2 F}{\partial^2 r} - \frac{2 \sin \theta \cos \theta}{r} \frac{\partial^2 F}{\partial r \partial \theta} + \frac{\sin^2 \theta}{r} \frac{\partial^2 F}{\partial^2 \theta} + \frac{\sin^2 \theta}{r} \frac{\partial F}{\partial r} + \frac{2 \sin \theta \cos \theta}{r^2} \frac{\partial F}{\partial \theta} \\ \frac{\partial^2 f}{\partial^2 y} &= \frac{\partial}{\partial y} \left(\sin \theta \frac{\partial F}{\partial r} + \frac{\cos \theta}{r} \frac{\partial F}{\partial \theta} \right) \\ &= \frac{\partial (\sin \theta)}{\partial y} \frac{\partial F}{\partial r} + \sin \theta \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial r} \right) + \frac{\partial}{\partial y} \left(\frac{\cos \theta}{r} \right) \frac{\partial F}{\partial \theta} + \frac{\cos \theta}{r} \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial \theta} \right) \\ &= \cos \theta \frac{\partial \theta}{\partial y} \frac{\partial F}{\partial r} + \sin \theta \left(\frac{\partial^2 F}{\partial^2 r} \sin \theta + \frac{\partial^2 F}{\partial r \partial \theta} \frac{\cos \theta}{r} \right) - \frac{2 \sin \theta \cos \theta}{r^2} \frac{\partial F}{\partial \theta} + \left(\frac{\partial^2 F}{\partial r \partial \theta} \sin \theta + \frac{\partial^2 F \cos \theta}{\partial^2 \theta r} \right) \frac{\cos \theta}{r} \\ &= \frac{\cos^2 \theta}{r} \frac{\partial F}{\partial r} + \sin^2 \theta \frac{\partial^2 F}{\partial^2 r} + \frac{\sin \theta \cos \theta}{r} \frac{\partial^2 F}{\partial r \partial \theta} - \frac{2 \sin \theta \cos \theta}{r^2} \frac{\partial F}{\partial \theta} + \frac{\sin \theta \cos \theta}{r} \frac{\partial^2 F}{\partial r \partial \theta} + \frac{\cos^2 \theta}{r^2} \frac{\partial^2 F}{\partial^2 \theta} \end{aligned}$$

$$\begin{aligned} \Delta f(x, y) &= \frac{\partial^2 f}{\partial^2 x} + \frac{\partial^2 f}{\partial^2 y} \\ &= (\sin^2 \theta + \cos^2 \theta) \frac{\partial^2 F}{\partial^2 r} + \left(\frac{\sin^2 \theta}{r^2} + \frac{\cos^2 \theta}{r^2} \right) \frac{\partial^2 F}{\partial^2 \theta} + \left(\frac{\sin^2 \theta}{r} + \frac{\cos^2 \theta}{r} \right) \frac{\partial F}{\partial r} \\ &= \frac{\partial^2 F}{\partial^2 r} + \frac{1}{r} \frac{\partial F}{\partial r} + \frac{1}{r^2} \frac{\partial^2 F}{\partial^2 \theta} \end{aligned}$$

RESOLUTION EDP PAR PASSAGE EN COORDONNEES POLAIRES

Soit une fonction $f = f(x, y)$ définie sur $R^+ \times R - \{0, 0\}$. On supposera que f est de classe C^1 et vérifie l'EDP ci-dessous :

$$y \frac{\partial f}{\partial x} - x \frac{\partial f}{\partial y} = yx + y^3/x \quad (1)$$

Pour résoudre cette EDP, nous allons utiliser le passage en coordonnées polaires défini par $(x, y) = (r \cos \theta, r \sin \theta)$.

Soit $F(r, \theta) = f(r \cos \theta, r \sin \theta)$ et en réutilisant les résultats du calcul sur le gradient (exercice précédent),

1. Calculer l'EDP vérifiée par F
2. Trouver une primitive de la fonction $\tan \theta$
3. En déduire la solution de (1)

$$1. F(r, \theta) = f(r \cos \theta, r \sin \theta)$$

$$y \frac{\partial f}{\partial x} - x \frac{\partial f}{\partial y} = yx + \frac{y^3}{x} \quad \text{avec } (x, y) \neq (0, 0)$$

$$\stackrel{\text{ex01}}{\Leftrightarrow} r \sin \theta \left(\cos \theta \frac{\partial F}{\partial r} - \frac{\sin \theta}{r} \frac{\partial F}{\partial \theta} \right) - r \cos \theta \left(\sin \theta \frac{\partial F}{\partial r} + \cos \theta \frac{\partial F}{\partial \theta} \right) = r^2 \sin \theta \cos \theta + r^2 \frac{\sin^3 \theta}{\cos \theta}$$

$$\Leftrightarrow -(\sin^2 \theta + \cos^2 \theta) \frac{\partial F}{\partial \theta} = \frac{\sin \theta (\cos^2 \theta + \sin^2 \theta)}{\cos \theta} r^2$$

$$\Leftrightarrow -\frac{\partial F}{\partial \theta} = r^2 \frac{\sin \theta}{\cos \theta}$$

$$\Leftrightarrow -\frac{\partial F}{\partial \theta} = \tan \theta r^2. \quad \text{avec } r \neq 0 \text{ et } \theta \in]0, 2\pi[$$

$$2. F(r, \theta) = -r^2 \int \tan \theta d\theta + C(r) \\ = r^2 \ln |\cos \theta| + C(r)$$

$$3. f(x, y) = F(r, \theta) \\ = (x^2 + y^2) \ln \left| \frac{x}{\sqrt{x^2 + y^2}} \right| + C(\sqrt{x^2 + y^2}) \quad \text{avec } (x, y) \neq (0, 0)$$

ENCORE UN CHANGEMENT DE VARIABLE

En supposant que f est de classe C^1 , résoudre l'EDP ci-dessous :

$$x \frac{\partial f}{\partial x} - y \frac{\partial f}{\partial y} = x - y$$

En utilisant le changement variable : $u = xy$ et $v = x + y$

$$f(x, y) = F(u, v) = F(xy, x+y)$$

$$\frac{\partial u}{\partial x} = y, \quad \frac{\partial u}{\partial y} = x, \quad \frac{\partial v}{\partial x} = 1, \quad \frac{\partial v}{\partial y} = 1$$

$$\frac{\partial f}{\partial x} = \frac{\partial F}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial F}{\partial v} \frac{\partial v}{\partial x} = y \frac{\partial F}{\partial u} + \frac{\partial F}{\partial v}$$

$$\frac{\partial f}{\partial y} = \frac{\partial F}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial F}{\partial v} \frac{\partial v}{\partial y} = x \frac{\partial F}{\partial u} + \frac{\partial F}{\partial v}$$

$$x \frac{\partial f}{\partial x} - y \frac{\partial f}{\partial y} = x \left(y \frac{\partial F}{\partial u} + \frac{\partial F}{\partial v} \right) - y \left(x \frac{\partial F}{\partial u} + \frac{\partial F}{\partial v} \right) \\ = x \frac{\partial F}{\partial v} - y \frac{\partial F}{\partial v} \\ = (x - y) \frac{\partial F}{\partial v}$$

$$\text{Alors l'équation est } (x - y) \frac{\partial F}{\partial v} = x - y$$

$$\text{Alors } \frac{\partial F}{\partial v} = 1$$

$$F(u, v) = v + H(u)$$

$$f(x, y) = (x + y) + H(xy)$$

UN DERNIER CHANGEMENT DE VARIABLE

En supposant que f est de classe C^1 , résoudre l'EDP ci-dessous

$$\frac{\partial f}{\partial x} - 3 \frac{\partial f}{\partial y} = 0$$

Faites le bon changement de variable.

On effectue un changement de variable

$$\begin{cases} u = \alpha x + \beta y \\ v = \gamma x + \delta y \end{cases}, \text{ avec } \alpha \delta \neq \gamma \beta$$

et on pose $f(x, y) = F(u, v)$

Par la règle de la chaîne, on obtient

$$\begin{aligned} \frac{\partial f}{\partial x} &= \frac{\partial F}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial F}{\partial v} \frac{\partial v}{\partial x} \\ &= \alpha \frac{\partial F}{\partial u} + \gamma \frac{\partial F}{\partial v} \\ \frac{\partial f}{\partial y} &= \frac{\partial F}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial F}{\partial v} \frac{\partial v}{\partial y} \\ &= \beta \frac{\partial F}{\partial u} + \delta \frac{\partial F}{\partial v} \end{aligned}$$

En remplaçant dans l'équation, il vient

$$\frac{\partial f}{\partial x} - 3 \frac{\partial f}{\partial y} = 0$$

$$\Leftrightarrow \alpha \frac{\partial F}{\partial u} + \gamma \frac{\partial F}{\partial v} - 3\beta \frac{\partial F}{\partial u} - 3\delta \frac{\partial F}{\partial v} = 0$$

$$\Leftrightarrow (\alpha - 3\beta) \frac{\partial F}{\partial u} + (\gamma - 3\delta) \frac{\partial F}{\partial v} = 0 \quad (i)$$

On choisit le changement de variable de façon à simplifier l'équation. En imposant

$$\begin{cases} \alpha = 3 \\ \beta = 1 \end{cases} + \begin{cases} \gamma = 0 \\ \delta = 1 \end{cases} \text{ et } \begin{cases} u = 3x + y \\ v = y \end{cases}$$

2) Résoudre l'équation (1)

D'après le changement de variable question 1)

$(\alpha - 3\beta) = 0$ et $\gamma - 3\delta = -3$. L'équation (i) se réduit alors à $\frac{\partial F}{\partial v} = 0$

Il existe donc une fonction arbitraire H de classe C^1

telle que $F(u, v) = H(u)$. Ainsi, la solution générale est $f(x, y) = H(3x + y)$

Supposons que
$$\begin{cases} f(x,y) = 0 & , \text{ si } xy = 0 \\ f(x,y) = 1 & , \text{ si } xy \neq 0 \end{cases}$$

Calculer $\frac{\partial f}{\partial x}(0,0)$ et $\frac{\partial f}{\partial y}(0,0)$.

Existence.

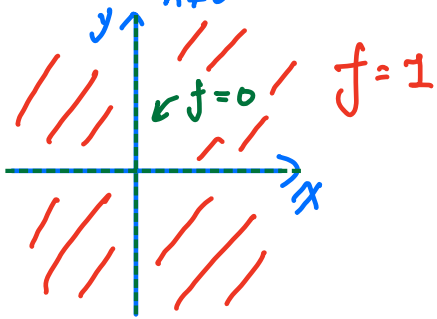
f est-elle continue en $(0,0)$?

Dérivées partielles

$$\frac{\partial f}{\partial x}(0,0) = \lim_{h \rightarrow 0} \frac{f(h,0) - f(0,0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{0}{h} = 0$$

$$\frac{\partial f}{\partial y}(0,0) = \lim_{h \rightarrow 0} \frac{f(0,h) - f(0,0)}{h} = 0$$



f n'est pas continue en $(0,0)$

$$f(0,0) = 0$$

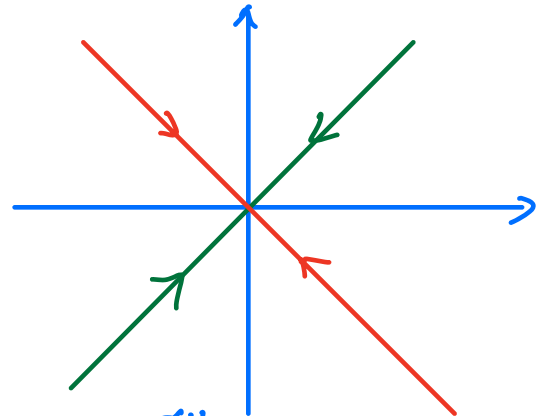
$$f(x,y) = \frac{xy}{x^2+y^2}, \quad (x,y) \neq 0$$

Continue ?

$$\begin{cases} f(0,0) = 0 \\ f(x,y) = \frac{xy}{x^2+y^2}, \text{ avec } (x,y) \neq 0 \end{cases}$$

$$\frac{\partial f}{\partial x}(0,0) = \lim_{h \rightarrow 0} \frac{f(h,0) - f(0,0)}{h} = 0$$

$$\frac{\partial f}{\partial x}(0,0) = \lim_{h \rightarrow 0} \frac{f(0,h) - f(0,0)}{h} = 0$$



$$f(x,y) = \frac{xy}{x^2+y^2}$$

$$f(x,x) = \frac{x^2}{x^2+x^2} = \frac{1}{2}, \quad x \neq 0$$

$$f(x,-x) = \frac{-x^2}{x^2+x^2} = -\frac{1}{2}, \quad x \neq 0.$$